

Lecture 13: Kolmogorov Smirnov Test & Power of Tests

S. Massa, Department of Statistics, University of Oxford

2 February 2016

An example

Suppose you are given the following 100 observations.

-0.16	-0.68	-0.32	-0.85	0.89	-2.28	0.63	0.41	0.15	0.74
1.30	-0.13	0.80	-0.75	0.28	-1.00	0.14	-1.38	-0.04	-0.25
-0.17	1.29	0.47	-1.23	0.21	-0.04	0.07	-0.08	0.32	-0.17
0.13	-1.94	0.78	0.19	-0.12	-0.19	0.76	-1.48	-0.01	0.20
-1.97	-0.37	3.08	-0.40	0.80	0.01	1.32	-0.47	2.29	-0.26
-1.52	-0.06	-1.02	1.06	0.60	1.15	1.92	-0.06	-0.19	0.67
0.29	0.58	0.02	2.18	-0.04	-0.13	-0.79	-1.28	-1.41	-0.23
0.65	-0.26	-0.17	-1.53	-1.69	-1.60	0.09	-1.11	0.30	0.71
-0.88	-0.03	0.56	-3.68	2.40	0.62	0.52	-1.25	0.85	-0.09
-0.23	-1.16	0.22	-1.68	0.50	-0.35	-0.35	-0.33	-0.24	0.25

Do they come from $N(0,1)$?

The Kolmogorov-Smirnov Test

Suppose that we have observations $X_1, \dots, X_n$, which we think come from a distribution P .

The **Kolmogorov-Smirnov Test** is used to test

$\mathbf{H}_0$: the samples come from P ,

against

$\mathbf{H}_1$: the samples do not come from P .

Cumulative Distribution Function and Empirical Distribution Function

The **cumulative distribution function** $F(x)$ of a random variable X , is

$$F(x) = \mathbb{P}(X \leq x).$$

The cumulative distribution function **uniquely characterizes** a probability distribution.

Given observations $x_1, \dots, x_n$ the **empirical distribution function** $F_{\text{obs}}(x)$ gives the proportion of the data that lies below x ,

$$F_{\text{obs}}(x) = \frac{\text{\#observations below } x}{\text{\#observations}}.$$

If we order the observations $y_1 \leq y_2 \leq \dots \leq y_n$, then

$$F_{\text{obs}}(y_i) = \frac{i}{n}.$$

The Kolmogorov-Smirnov statistic

We want to compare the empirical distribution function of the data, F_{obs} , with the cumulative distribution function associated with the null hypothesis, F_{exp} (expected CDF).

The Kolmogorov-Smirnov statistic is

$$D_n = \max_x |F_{\text{exp}}(x) - F_{\text{obs}}(x)|.$$

The practical approach

In practice, **order the data**:

-3.68	-2.28	-1.97	-1.94	-1.69	-1.68	-1.60	-1.53	-1.52	-1.48
-1.41	-1.38	-1.28	-1.25	-1.23	-1.16	-1.11	-1.02	-1.00	-0.88
-0.85	-0.79	-0.75	-0.68	-0.47	-0.40	-0.37	-0.35	-0.35	-0.33
-0.32	-0.26	-0.26	-0.25	-0.24	-0.23	-0.23	-0.19	-0.19	-0.17
-0.17	-0.17	-0.16	-0.13	-0.13	-0.12	-0.09	-0.08	-0.06	-0.06
-0.04	-0.04	-0.04	-0.03	-0.01	0.01	0.02	0.07	0.09	0.13
0.14	0.15	0.19	0.20	0.21	0.22	0.25	0.28	0.29	0.30
0.32	0.41	0.47	0.50	0.52	0.56	0.58	0.60	0.62	0.63
0.65	0.67	0.71	0.74	0.76	0.78	0.80	0.80	0.85	0.89
1.06	1.15	1.29	1.30	1.32	1.92	2.18	2.29	2.40	3.08

Then **compute the empirical distribution function**:

$$F_{\text{obs}}(-3.68) = \frac{1}{100}, \quad F_{\text{obs}}(-2.28) = \frac{2}{100}, \dots, \quad F_{\text{obs}}(3.08) = 1.$$

If our data is ordered, x_1 being the least and x_n being the largest, then

$$F_{\text{obs}}(x_i) = \frac{i}{100}.$$

The practical approach

0.000	0.011	0.024	0.026	0.045	0.047	0.055	0.064	0.064	0.070
0.080	0.084	0.101	0.107	0.110	0.123	0.133	0.154	0.158	0.189
0.198	0.215	0.226	0.249	0.321	0.343	0.356	0.362	0.363	0.369
0.375	0.396	0.399	0.400	0.407	0.409	0.410	0.423	0.425	0.432
0.432	0.434	0.437	0.447	0.449	0.453	0.464	0.468	0.476	0.477
0.484	0.484	0.485	0.490	0.496	0.505	0.508	0.526	0.535	0.553
0.557	0.560	0.577	0.577	0.582	0.588	0.597	0.610	0.614	0.617
0.627	0.658	0.680	0.692	0.698	0.711	0.720	0.727	0.732	0.735
0.743	0.748	0.761	0.771	0.777	0.783	0.788	0.789	0.803	0.812
0.854	0.874	0.902	0.903	0.907	0.973	0.985	0.989	0.992	0.999

For each observation x_i compute $F_{\text{exp}}(x_i) = P(Z \leq x_i)$.

In this case the expected distribution function is standard normal so use the normal table.

The practical approach

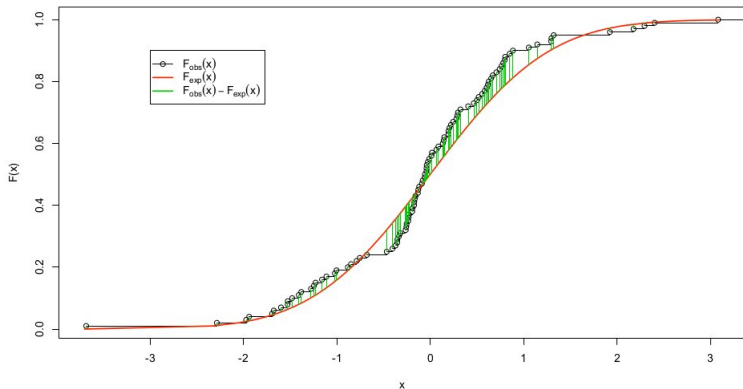
Compute the **absolute differences** between the entries in the two tables.

The Kolmogorov-Smirnov statistic $D_n = 0.092$ is the maximum shown here in blue.

0.010	0.009	0.006	0.014	0.004	0.014	0.015	0.017	0.026	0.031
0.031	0.036	0.030	0.034	0.041	0.037	0.037	0.026	0.031	0.011
0.012	0.005	0.003	0.008	0.069	0.085	0.086	0.083	0.073	0.071
0.064	0.077	0.067	0.061	0.055	0.049	0.039	0.045	0.035	0.033
0.023	0.013	0.006	0.008	0.002	0.008	0.006	0.012	0.014	0.024
0.026	0.036	0.046	0.052	0.054	0.056	0.062	0.052	0.054	0.048
0.054	0.060	0.055	0.061	0.067	0.073	0.071	0.070	0.076	0.082
0.084	0.061	0.049	0.049	0.052	0.048	0.051	0.054	0.058	0.064
0.068	0.071	0.069	0.070	0.074	0.078	0.082	0.092	0.088	0.087
0.055	0.045	0.029	0.037	0.043	0.013	0.015	0.009	0.002	0.001

The Kolmogorov-Smirnov Statistic

We have calculated the maximum absolute distance between the expected and observed distribution functions, in green in the plot below.



What is the Critical Value?

- ▶ At the 95% level the critical value is approximately given by

$$D_{\text{crit},0.05} = \frac{1.36}{\sqrt{n}}.$$

- ▶ Here we have a sample size of $n = 100$ so $D_{\text{crit}} = 0.136$.
- ▶ Since $0.092 < 0.136$ **do not reject** the null hypothesis.

Kolmogorow Smirnov for Two Samples

Given two samples, test if their distributions are the same.

Compute the observed cumulative distribution functions of the two samples and compute their maximum difference.

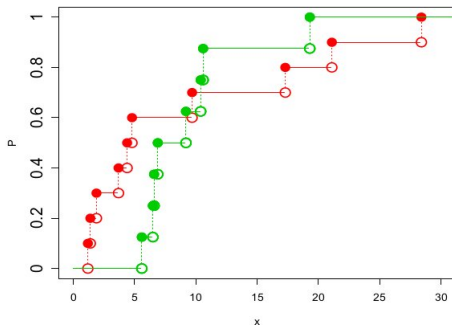
$X : 1.2, 1.4, 1.9, 3.7, 4.4, 4.8, 9.7, 17.3, 21.1, 28.4$

$Y : 5.6, 6.5, 6.6, 6.9, 9.2, 10.4, 10.6, 19.3.$

- We sort the combined sample, in order to compute the empirical cdf's:

	1.2	1.4	1.9	3.7	4.4	4.8	5.6	6.5	6.6	6.9
F_x	0.1	0.2	0.3	0.4	0.5	0.6	0.6	0.6	0.6	0.6
F_y	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.4	0.5
	9.2	9.7	10.4	10.6	17.3	19.3	21.1	28.4		
F_x	0.6	0.7	0.7	0.7	0.8	0.8	0.9	1.0		
F_y	0.6	0.6	0.8	0.9	0.9	1.0	1.0	1.0		

Kolmogorow Smirnov for Two Samples



The Kolmogorov-Smirnov statistic is again the maximum absolute difference of the two observed distribution functions. Here

$$D_n = 0.6.$$

Kolmogorow Smirnov for Two Samples

For two samples, the 95% critical value can be approximated by the formula:

$$D_{\text{crit},0.05} = 1.36 \sqrt{\frac{1}{n_x} + \frac{1}{n_y}}.$$

In our case $n_x = 10$ and $n_y = 8$ and thus $D_{\text{crit}} = 0.645$.

So we retain the null hypothesis.